

Hexadecimal · IG · Answers

U04 · 24 written + 6 pen-first = 30 · Student copy

Authored schemes, informed by published question shapes. Accept equivalent clear wording and valid working. Do not credit contradictions. Hex letter case and spaces between binary groups do not change a value.

1. Conversions [6]

- (a) **B6** [1]. (b) **00000111** [1]; eight bits required.
(c) $230 \div 16 = 14$ remainder 6 (or $230 = 14 \times 16 + 6$) [1]; **E6** [1].
(d) $13 \times 256 + 0 \times 16 + 8$ [1]; **3336** [1]. Accept equivalent valid conversion working; do not award a working mark for an unsupported answer alone.

2. Why hexadecimal? [2]

- (a) Fewer symbols to read/write, or easier human reading/transcription [1]. “Less storage” is not a benefit of changing the displayed notation. (b) Four bits have $2^4 = 16$ patterns, matching the 16 available hex digits [1].

3. Authored colour bytes [6]

- (a) **0100 1110**; (b) **1010 0010**; (c) **0000 1001**. One mark per correctly positioned four-bit nibble. All 8 positions are requested; zero nibbles matter. A wrong nibble does not remove credit for the other correct nibble.

4. Two other uses [2]

Any two distinct valid examples, 1 each, Max 2: memory addresses; memory dumps; hexadecimal error codes; IPv6 addresses; machine-code bytes displayed in a debugger. Do not accept the excluded colour/MAC examples, “computers” alone, or a statement that denies the use.

5. Value and width [8]

- (a) **5A3E:5A** [1], **3E** [1], in the correct positions.
(b) $351 \div 16 = 21$ r 15; $21 \div 16 = 1$ r 5; $1 \div 16 = 0$ r 1 [1]; **15F** [1]. Equivalent place-value/binary route accepted.
(c) $128 + 32 + 8 + 4 + 1$ [1]; **173** [1].
(d) Yes, both denote 3 [1]; **00000011** for an 8-bit register [1].

6. Pen-first translator [6]

- (a) Group from the right and pad with zeroes on the left to a multiple of four [1]; translate each group using the nibble table and preserve order [1]. A clear numbered method or flowchart is acceptable. Equivalent correct algorithms accepted.
(b) **0001 1011 1001** [1]; **1B9** [1]. (c) **0000 1111** [1]; the supplied first hex digit is 0, which expands to four zero bits under the stated contract [1].

Unmarked RGB extension: channels **40, 100, 180**; new blue $196 = C4$, giving **#2864C4**. Red and green stay unchanged.

Optional coding challenge: no extra marks. Require both directions, correct left padding, four output bits per supplied hex digit, preserved leading zero nibbles, lowercase hex acceptance and rejection of invalid/empty input. The 13 local examples are not a proof. Check the `no-int()/hex()/bin()` rule by reading the converter functions. A nibble lookup table supplies a model approach.